3.2.R1 (1/1 point)

We run a linear regression and the slope estimate is 0.5 with standard error of 0.2. What is the largest value of b for which we would NOT reject the null hypothesis that $\beta_1 = b$? (Assume we are using the 5% significance level)

0.9

0.9

Answer: 0.9

EXPLANATION

The 95% confidence interval $\hat{\beta}_1 \pm 2S.E.(\hat{\beta}_1)$ contains all parameter values that would not be rejected at a 5% significance level.

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3.2.R2 (1/1 point)

Which of the following indicates a fairly strong relationship between X and Y?

- [x] $R^2 = 0.9$ ✔
- [ ] The p-value for the null hypothesis $\beta_1 = 0$ is 0.0001
- [ ] The t-statistic for the null hypothesis $\beta_1 = 0$ is 30

EXPLANATION

The R^2 is the correlation between the two variables and measures how closely they are associated. The p value and t statistic merely measure how strong is the evidence that there is a nonzero association. Even a weak effect can be extremely significant given enough data.

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